

Human-Agentic Framework for Scientific Research and Publication Genesis

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Abstract

The development of large language models has opened new opportunities for preparing scientific publications, analyzing literature, formulating hypotheses, constructing models, programming, and processing research results. At the same time, using a language model primarily as a generator of scientific text does not address a fundamental problem: a scientific publication should be the outcome of a research process rather than the result of sequential generation of plausible text. This paper proposes Human-Agentic Article Genesis (HAAG), a human-agentic, prompt-native, no-code framework for organizing interactive scientific research and creating scientific publications. The main idea of HAAG is to shift the central object of control from the article text to the research trajectory. Specialized LLM agents explore the space of problems, hypotheses, models, methods, and evidence, while the researcher defines the scientific intent and controls transitions capable of substantially changing the content and direction of the research. The framework introduces a research-state model, a system of logical primitives, a society of specialized agents, and the `HUMAN` operator, which formalizes the researcher's participation in scientifically consequential decisions. HAAG incorporates mechanisms for hypothesis falsification, experiment preregistration, numerical auditing, claim provenance, source verification, handling of negative results, generation of scientific figures, and preparation of a reproducibility package. Unlike one-shot generation of an article from a given topic, HAAG implements the *think–challenge–verify–decide–run–audit–transition–write* cycle, in which text is generated only from an accepted research state. Practical application of the framework throughout a complete theoretical and modeling research workflow was used to refine the architecture of HAAG v1.1. The results demonstrate that HAAG can be viewed as a human-readable domain-specific language for managing scientific research, in which natural language provides semantics, logical primitives provide control, LLM agents perform distributed research functions, and the human provides scientific intent and shapes the research trajectory.

Keywords: large language models, LLM, agentic systems, human-agent interaction, scientific research, scientific publication, HAAG, reproducibility, traceability, generative artificial intelligence.

1. Introduction

In recent years, large language models (Large Language Models, LLMs) have moved beyond their use as tools for text generation and editing and have become increasingly integrated into various stages of scientific activity. They are used for searching and synthesizing literature, formulating research questions and hypotheses, constructing mathematical and conceptual models, generating program code, planning experiments, analyzing data, interpreting results, and preparing scientific publications. Recent reviews already consider LLMs not merely as scientific writing tools but as components of a

broader cycle of scientific inquiry, from hypothesis formulation to hypothesis testing and the generation of new knowledge [1].

Particular attention has been given to the use of LLMs for scientific hypothesis generation. A systematic review by Alkan et al. demonstrates a transition from simple prompting toward more sophisticated schemes involving structured reasoning, novelty search, external sources, and human–model interaction. At the same time, the authors note that evaluating the novelty and scientific validity of generated hypotheses, as well as organizing productive human–AI collaboration, remain open problems [2].

The next stage in this development has been the emergence of LLM-based scientific agents – systems in which a language model is assigned a specialized role, memory, access to external tools, and the ability to perform sequences of research actions. In the review by Ren et al., such agents are considered a distinct class of systems capable of supporting hypothesis formulation, experiment planning, data analysis, and modeling. The authors associate further development of scientific agents with the need to integrate domain knowledge, tools, and mechanisms for verification and reproducibility [3].

This trend has naturally led to the emergence of systems aimed not at a single operation but at a substantial part, or even the entire cycle, of scientific research. One representative example is The AI Scientist, proposed by Lu et al. The system generates research ideas, creates and modifies program code, conducts experiments, visualizes results, produces a scientific paper, and performs automated peer review. The authors position this approach as a step toward fully automated open-ended scientific discovery and demonstrate it on machine-learning tasks [4].

Another approach is implemented in Agent Laboratory. Here, the research process is divided into three main stages: literature review, experimentation, and report writing. Importantly, the system allows human participation and feedback at individual stages. The authors' experimental evaluation showed that such human involvement improved the quality of the final result, providing a substantial argument in favor not only of autonomous but also of hybrid human-agent schemes for organizing scientific research [5].

The collaborative model is even more explicit in the AI co-scientist system developed by Google Research. It is designed as a multi-agent system for generating and refining scientific hypotheses and research proposals. In contrast to the concept of fully automated science, its developers explicitly position it as a tool for collaboration with scientists rather than as a means of removing them from the research process. The researcher specifies a scientific goal in natural language, after which the system can propose hypotheses, connect them with relevant literature, and formulate possible experimental directions [6].

Thus, contemporary research reveals a gradual transition

LLM as Writer→LLM as Assistant→Scientific Agent→

→Multi-Agent Research System→AI-supported Research Process

This evolution changes the formulation of the problem itself. Whereas at an earlier stage the main question concerned the quality of scientific text that a language model could generate, an increasingly

important question is now how to organize a reliable research process involving humans, LLM agents, external sources, and computational tools.

A separate aspect of the human-agent organization of scientific activity concerns preserving the role of the human as the subject of scientific reasoning and authorship. The HAAR (Human Authorship Assessment and Reasoning) framework proposes an approach for assessing an author's command of the logic of scientific research by reconstructing semantic relationships among problem formulation, hypotheses, methods, results, and conclusions [7]. In this sense, HAAR and the HAAG framework proposed in this paper address complementary aspects of human-agent science: HAAG formalizes the process of conducting research, whereas HAAR addresses the assessment of substantive authorial control over scientific reasoning.

At the same time, the current literature reveals several problems associated with this transition. Automatically generating a plausible hypothesis does not guarantee its novelty; performing a computational experiment does not guarantee that the problem has been formulated correctly; the existence of a source does not mean that it supports a particular claim; and a formally complete article does not necessarily make it possible to reconstruct the sequence of decisions that led to its results. A broad review of the role of LLMs in the scientific method similarly emphasizes the need to integrate such systems with human scientific goals and to use explicit procedures for evaluating their outputs [8].

Thus, alongside the problem of automating individual research operations, another task arises—the organization of the research process itself. It is necessary to determine who defines the next research step and on what grounds; when a decision can be delegated to an agent and when it should be referred to a human; how alternative and rejected hypotheses should be preserved; how preliminary results should be distinguished from verified ones; how negative results should be recorded; how claims should be linked to evidence; and, ultimately, how the trajectory leading from the initial topic to the final publication can be reconstructed.

This problem constitutes the starting point of the Human-Agentic Article Genesis (HAAG) framework proposed in this paper.

HAAG is not conceived as a universal prompt of the form “write a scientific article on a given topic,” but as a human-agentic framework for organizing interactive scientific research. Within it, a set of specialized LLM agents explores the space of problems, hypotheses, models, methods, and evidence, while the researcher participates in transitions that have substantial scientific consequences. The publication emerges as the outcome of a traceable human-agent research trajectory; therefore, the central object of HAAG is not the article text but the research process.

The core principle of HAAG is formulated as follows:

Agents explore the research space; the human shapes the research trajectory.

This principle means that the agentic system can expand the space of considered alternatives, generate and criticize hypotheses, construct models, propose experiments, and verify results, while scientifically consequential transitions within the research trajectory remain explicit and under the researcher's control. For this purpose, HAAG formalizes the human as a separate `HUMAN` operator

rather than merely treating the human as the source of the initial prompt or the editor of the final text. This makes it possible to move from the scheme

Topic→Prompt→Publication

to the scheme

**Topic + Profile→ Research Program→ Human-Agent Research→ Trajectory→
→Verified Results → Verified Publication + Reproducibility Package**

The formal HAAG specification maintains the research state, a specialized society of agents, logical primitives for controlling the trajectory, branching and return to previous states, source verification, experiment preregistration and freezing, numerical auditing, claim provenance, handling of negative results, generation of scientific figures, and preparation of a reproducibility package. Importantly, every consequential transition must be attributable to an agent action, verified evidence, a mathematical or logical rule, a numerical result, or a human decision.

Aim and Contribution of the Work

The **aim of this work** is to develop and formalize a human-agent approach to organizing the complete scientific research cycle, in which LLM agents are used not merely to generate scientific text but as specialized participants in a controlled, verifiable, and traceable research trajectory.

The scientific and methodological contribution of this work consists in formalizing HAAG as a system that combines: **(1)** a scientific research-state model; **(2)** a prompt-native system of logical primitives for controlling the research process; **(3)** a society of functionally specialized LLM agents; **(4)** the `HUMAN` operator for scientifically consequential transitions; **(5)** `FORK`, `MERGE`, and `BACKTRACK` mechanisms for a nonlinear research trajectory; **(6)** the `PREREGISTER-FREEZE-RUN-AUDIT` and `SOURCE-VERIFY-CITE` cycles; **(7)** provenance of major scientific claims; **(8)** separation of the research, writing, and artifact layers; and **(9)** generation of a publication together with a reproducibility package. The corresponding set of primitives and agent functions is formally defined in the HAAG specification available in the (<https://github.com/DmytroLande/HAAG/>).

The fundamental distinction of the proposed approach can be expressed through a change in the primary question posed to the system. Instead of

What text should be generated next?

HAAG asks

What scientifically justified research transition should occur next?

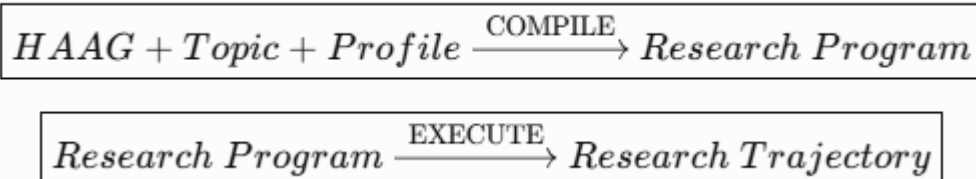
Only after the corresponding transitions have been performed and verified is the publication text generated. This rule is explicitly embedded in the HAAG specification (<https://github.com/DmytroLande/HAAG/tree/main/Framework>).

2. HAAG Concept

The Human-Agentic Article Genesis (HAAG) concept is based on a transition from using large language models as tools for direct text generation to integrating them into a controlled process of designing and conducting scientific research. In this setting, a scientific publication remains the final product of the system, but it is not the direct result of the initial query. The publication is produced at the final stages of the research trajectory on the basis of results that have been obtained, verified, and properly recorded. This approach reflects a fundamental principle of HAAG: the central object is the research process, while the article text is treated as one of its output artifacts.

The initial interface of the framework is intentionally minimal. To initiate a study, the user only needs to specify a topic (`TOPIC`). Additional parameters may include the publication type, domain, language, desired length, target venue, availability of real data, visualization requirements, and requirements for a reproducibility repository. If these characteristics are not specified in advance, the system may determine them while constructing the research program, provided that the corresponding decisions do not alter the scientific content of the work. At the same time, HAAG requires human involvement whenever a choice affects the problem formulation, evidence base, research method, experimental design, interpretation of results, or final form of the scientific work.

At the conceptual level, the operation of HAAG can therefore be represented as a two-stage transformation:



The `COMPILE` operation does not mean generating a finished text; rather, it constructs a domain-specific research program. It defines the initial problem structure, research questions, possible hypotheses, agent composition, required sources and tools, potential verification procedures, experimental actions, and points at which human decisions may be required. The `EXECUTE` operation, in turn, implements this program as a sequence of research states, with transitions between them producing new hypotheses, evidence, models, results, rejected alternatives, and other scientific artifacts. This distinction between compiling a research program and executing it is embedded in the HAAG architecture.

In this sense, HAAG can be interpreted as a compiler that transforms research intent into an interactive scientific research program. This interpretation differs fundamentally from the universal-prompt model, in which a topic is transformed directly into text. In HAAG, an intermediate layer—a formalized research program—is introduced between the initial research intent and the publication. Execution of this program may be nonlinear and may involve alternative branches, returns to previous states, rejection of hypotheses, and reformulation of individual research elements. For this purpose, the framework specification includes, among others, the primitives `FORK`, `MERGE`, `BACKTRACK`, `VERIFY`, `ACCEPT`, `REJECT`, `PREREGISTER`, `RUN`, and `AUDIT`.

Thus, instead of the direct scheme

Topic→ Prompt→Text

HAAG implements the more elaborate sequence

Topic→ Research Program→ Research Trajectory→ Verified Results→ Publication.

The concept of a **research trajectory** is fundamental here. Scientific research rarely unfolds as a predetermined linear sequence of operations. Competing explanations, alternative mathematical models, conflicting sources, negative experimental results, or the need to revisit an earlier assumption may arise during the research process. HAAG does not attempt to conceal this nonlinearity behind a coherent final text. On the contrary, branching paths, rejected directions, and failed experiments are treated as part of the research process and are preserved in its trace (`TRACE`).

The interaction between the human and the agentic system within this trajectory is governed by the basic HAAG principle:

<i>Agents explore the research space; Human shapes the research trajectory.</i>
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This principle establishes a functional distinction between the research space and the research trajectory. The research space comprises the set of potentially admissible research questions, hypotheses, counterhypotheses, models, methods, sources, experimental designs, and interpretations. Specialized agents can explore this space in parallel: `Explorer` identifies problems and gaps; `Theorist` formulates hypotheses and models; `Contrarian` proposes alternative explanations; `Falsifier` searches for ways to refute hypotheses; `Experimentalist` designs tests; and `Reviewer`, `Source Auditor`, and `Numerical Auditor` critically evaluate the resulting findings, sources, and computations.

The research trajectory, unlike the space of possibilities, is a specific sequence of scientifically consequential transitions. It determines which of the generated alternatives are accepted for further investigation, which are rejected, which require additional verification, and when a return to a previous state is necessary. Therefore, increasing agent autonomy in HAAG does not imply automatically delegating all scientific decisions to agents. Agent autonomy is used primarily to expand the explored research space and reduce the likelihood that a potentially important alternative remains unexamined.

The role of the human is formalized not as external observation of the system's operation but as an integral part of the state-transition mechanism itself. For this purpose, HAAG introduces the `HUMAN` operator. It is invoked, in particular, before choosing among substantively different hypotheses, changing a model, choosing between real and synthetic data, modifying an experiment after obtaining a result, changing a metric or criterion, making a strong novelty claim, interpreting an ambiguous negative result, or performing the final synthesis of the work. At the same time, routine operations that do not alter the scientific trajectory do not require a separate decision by the researcher.

In general form, this principle can be represented as a controlled transition between states:

$$S_{t+1} = \begin{cases} F_A(S_t), & \text{if the transition is routine or unambiguously justified,} \\ F_H(S_t, d_H), & \text{if the transition has substantial scientific consequences,} \end{cases}$$

where S_t is the current research state, F_A is a transition generated by the agentic system on the basis of the available state, and F_H is a transition that depends on the researcher's decision d_H . This formalization does not replace the more detailed HAAG state model discussed below; rather, it captures the conceptual principle governing the division of functions between the human and the agentic system.

Thus, HAAG does not place agent autonomy in opposition to the researcher's scientific responsibility. Instead, they perform complementary functions: agents enable scalable exploration, criticism, verification, and comparison of multiple possible directions, while the human determines the scientifically meaningful trajectory through this space. As a result, the publication-generation process takes the form of a controlled human-agent evolution of the research state:

$$S_0 \xrightarrow{\text{Agents, Tools, Sources, HUMAN}} S_1 \longrightarrow \dots \longrightarrow S_n \rightarrow \text{Verified Scientific Publication.}$$

The general HAAG architecture, integrating research-program construction, its human-agent execution, verification of results, and generation of final scientific artifacts, is presented in **Fig. 1**.

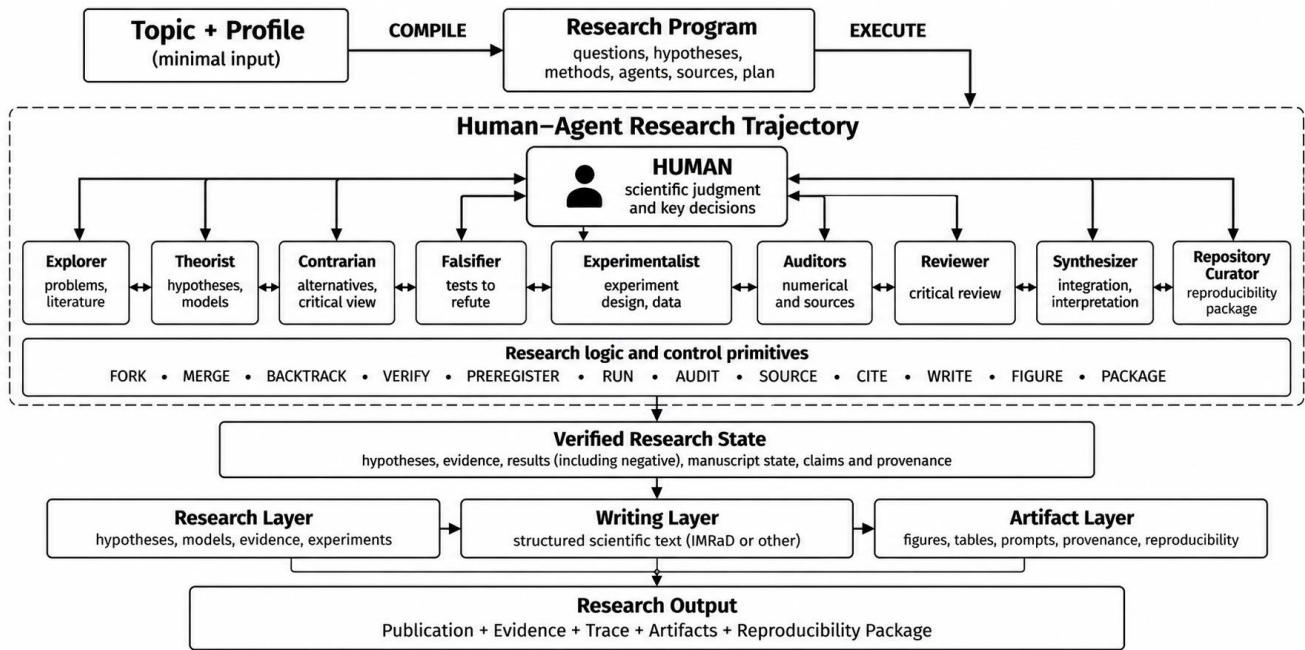


Fig. 1. Conceptual architecture of the human-agentic HAAG framework

As shown in Fig. 1, a publication in HAAG is generated not directly from the initial query, but from the verified results of a research trajectory governed by the interaction between specialized agents and the researcher.

Thus, the HAAG concept shifts the focus from automating scientific writing to organizing scientifically justified transitions between research states. In this model, the publication text does not determine the logic of the research; rather, it is generated from the verified final research state. This provides a conceptual basis for the subsequent formalization of the state (S_t) , transition mechanisms, agent society, the HUMAN operator, and traceability mechanisms, which are addressed in the following sections.

3. Formal Model of the Research Trajectory

To formally represent the research process in HAAG, a discrete state-evolution model is used. At each iteration (t) , the research process is characterized by the state

$$S_t = (T, H, Q, E, M, R, P, B, A, D, C, X, V, G),$$

where T defines the research topic and scope; H is the set of formulated and testable hypotheses; Q represents the research questions; E denotes the accumulated evidence and sources; M comprises the methods and mathematical and conceptual models; R represents the obtained results; P denotes the current state of the manuscript; B is the set of active, deferred, or rejected research branches; A represents the set of active agents; D contains the recorded decisions of the human and the agentic system; C represents scientific claims and their provenance; X denotes the history of the research process and the transitions performed; V represents the plan and generated scientific visualization artifacts; and G denotes the state of the reproducibility package. This state composition is explicitly defined in the formal HAAG specification.

The state S_t should not be regarded as a static collection of data, but rather as a formalized snapshot of the research at a particular stage of its execution. It simultaneously reflects the domain-specific content of the work, the current level of evidential support, accepted and alternative decisions, obtained results, and artifacts required for subsequent verification and reproduction. In this sense, the state carries the accumulated scientific context on the basis of which the next admissible research step is determined.

Accordingly, the research process in HAAG is represented not as a sequence of generated text fragments but as a trajectory through a state space:

$$S_0 \xrightarrow{\tau_0} S_1 \xrightarrow{\tau_1} S_2 \xrightarrow{\tau_2} \dots \xrightarrow{\tau_{n-1}} S_n,$$

Where $\tau_t : S_t \rightarrow S_{t+1}$ is a scientifically justified transition between two consecutive states. Such a transition may correspond to the formulation or rejection of a hypothesis, incorporation of a new source, construction or modification of a model, execution of an experiment, numerical verification, revision of an interpretation, selection of an alternative research branch, or another action that changes the research state.

In the general case, a transition can be represented by the operator

$$S_{t+1} = F_{\tau_t}(S_t, \omega_t),$$

where F_{τ_t} defines the transition rule and ω_t represents the basis for executing it. In HAAG, such a basis may be an action performed by a specialized agent, verified evidence or a source, a mathematical or logical rule, the result of an experiment or computation, or an explicit decision by the researcher. The formal specification requires every consequential transition to be associated with at least one of these types of justification.

This requirement can be expressed as a transition-provenance relation:

$$\text{Prov}(\tau_t) \subseteq \{Agent, Evidence, Rule, Result, HUMAN\}.$$

For a scientifically consequential transition, the set $\text{Prov}(\tau_t)$ must not be empty:

$$\text{Prov}(\tau_t) \neq \emptyset.$$

Thus, a transition between states must not only be performed but also be explained through a recorded justification. For example, accepting a hypothesis for further testing may be associated with the results produced by the agents and a HUMAN decision; a change in a numerical result may be associated with recomputation and auditing; and rejection of a claim may result from falsification or from a source that contradicts it.

This model extends the traditional notion of provenance. The object of traceability is not limited to the origin of a text fragment, source, or numerical value; it also includes the provenance of scientific decisions and transitions between research states. The history X, decisions D, claims C, alternative branches B, and other components of S_t collectively form a trace from which the logic of the research process can be reconstructed.

Consequently, the final state S_n characterizes not merely a completed manuscript P_n , but a collection of interconnected results, evidence, decisions, and artifacts. The generation of a scientific publication in HAAG can therefore be regarded as a projection of the verified research state:

$$P_{\text{final}} = \Pi_{\text{pub}}(S_n^{\text{verified}}),$$

where Π_{pub} is the operator that generates the publication from the verified part of the final state. In this formulation, the text represents the outcome of the research trajectory rather than replacing it. This provides a formal basis for implementing one of the central principles of HAAG: traceability not only of scientific claims, but also of the process through which they were obtained and the corresponding research decisions were made.

4. Human-Agent Organization of the Research Process

In HAAG, the research process is implemented through the interaction of specialized LLM agents and the researcher within a shared research trajectory. Instead of using a universal language model to sequentially perform heterogeneous operations, the framework decomposes the research process into functional roles. The basic agent society includes `Explorer`, `Theorist`, `Contrarian`, `Falsifier`, `Experimentalist`, `Numerical Auditor`, `Source Auditor`, `Figure Designer`, `Reviewer`, `Synthesizer`, `Repository Curator`, and `Research Orchestrator`. Each agent has a defined function, ranging from identifying research gaps and formulating hypotheses to verifying numerical results, sources, scientific claims, visualizations, and reproducibility materials.

The functional specialization of agents is used not only to distribute operations but also to organize a controlled internal scientific discussion. In particular, `Theorist` formulates hypotheses and models, while `Falsifier` attempts to identify conditions under which they can be refuted; `Contrarian` develops alternative explanations; `Experimentalist` proposes verification procedures; `Numerical Auditor` checks the correctness of quantitative results; and `Reviewer` evaluates the obtained results from the perspective of critical external assessment. Thus, different agents may perform not only complementary but also deliberately opposing research functions.

The corresponding cycle can be represented in a generalized form as

**Idea→ Hypothesis→ Counterhypothesis→ Falsification→
→ Experiment→ Audit→ Interpretation.**

This organization is fundamentally different from multi-agent generation of several variants of the same result. The HAAG agent society is used to expand the set of research alternatives under consideration, subject them to mutual criticism, and progressively reduce uncertainty regarding a scientifically justified research trajectory.

Coordination between agent autonomy and researcher responsibility is provided by the `HUMAN` operator, which in HAAG functions as a full-fledged state-transition operator. It is invoked when a choice among alternatives has substantial scientific consequences: when selecting among competing hypotheses, changing a model, determining the role of real or synthetic data, modifying an experiment after obtaining a result, changing metrics or criteria, making a strong novelty claim, interpreting ambiguous negative results, or performing the final synthesis of the work. Routine operations that do not alter the research trajectory may be performed autonomously by the agents.

In simplified form, this mechanism can be represented as

$$S_{t+1} = \begin{cases} F_A(S_t), & \text{for a routine or unambiguously justified transition,} \\ F_H(S_t, d_H), & \text{for a scientifically consequential transition,} \end{cases}$$

where F_A is the agent-based transition operator, F_H is the transition operator involving human participation, and d_H is the researcher's decision. At the same time, `HUMAN` is not limited to selecting one of the options proposed by the system. The researcher may accept, reject, modify, or combine

alternatives; narrow or broaden the problem formulation; return to a previous state; create a new branch; or terminate an existing one.

Human participation in HAAG should therefore be characterized more broadly than by the traditional *human-in-the-loop* paradigm. The human is directly integrated into the mechanism that shapes the research trajectory – *human-in-the-research-trajectory*. The agentic system explores the space of possibilities, while the researcher determines scientifically consequential directions for further progression.

The nonlinearity of this process is supported by a set of HAAG logical primitives. These include mechanisms for managing states and agents, conditions and loops, branching and returning (`FORK`, `MERGE`, `BACKTRACK`), verification (`VERIFY`, `ACCEPT`, `REJECT`), source handling (`SOURCE`, `CITE`), experimentation (`PREREGISTER`, `RUN`, `AUDIT`), text generation (`WRITE`, `REWRITE`), visualization (`FIGURE`, `FIGURE_PROMPT`), and final artifacts (`PACKAGE`, `EXPORT`). Together, they form a prompt-native means for describing and executing research logic.

Accordingly, the research process is naturally represented not as a linear chain but as a directed graph

$$\mathcal{G}_R = (\mathcal{S}, \mathcal{T}),$$

where $\mathcal{S}$ is the set of research states and $\mathcal{T} \subseteq \mathcal{S} \times \mathcal{S}$ is the set of admissible transitions between them. A particular research trajectory is a path through this graph, while the graph itself may contain alternative, deferred, rejected, and failed branches.

The nonlinear nature of the research process, the involvement of the `HUMAN` operator, branching, evaluation of alternatives, and preservation of rejected directions in `TRACE` are schematically illustrated in Fig. 2.

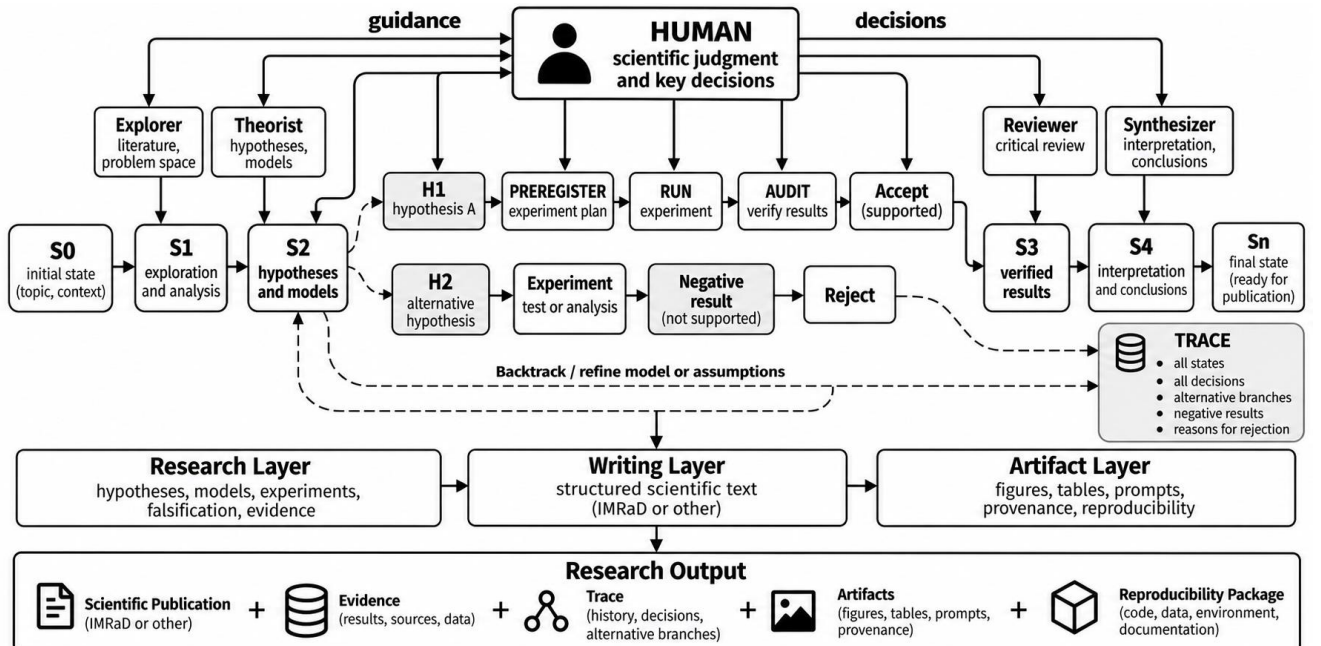


Fig. 2. Human-agent research trajectory in the HAAG state space

Fig. 2 emphasizes that the research trajectory represents only one of the possible paths through the state space. Rejected hypotheses, negative results, and failed experiments are not discarded but are preserved as elements of the traceable research history.

Importantly, the rejection of a hypothesis or the failure of an experiment does not imply that the corresponding branch is removed from the research history. HAAG treats negative and null results as valid scientific results, while a failed experiment is considered a distinct research state that must be recorded together with the cause of failure, its implications for scientific claims, and the admissible subsequent transitions.

Thus, `TRACE` preserves not only the history of the path that led to the final result but also information about alternatives that were examined and rejected. This integrates the agent society, the `HUMAN` operator, and the branching mechanisms into a unified model: ***agents generate and evaluate alternatives, the human controls scientifically consequential transitions, and the research graph preserves the provenance of both accepted and rejected decisions.***

5. Verification of Results, Publication Generation, and Reproducibility

One of the fundamental properties of HAAG is the separation of the processes of obtaining a scientific result, verifying it, and subsequently presenting it in the form of a publication. The framework is based on the premise that the completeness of a text cannot serve as a criterion for the completeness of a study. A scientific result must first be obtained, verified, and made traceable, and only then incorporated into the manuscript.

For computational and modeling experiments, HAAG uses the cycle

PREREGISTER → FREEZE → RUN → AUDIT.

At the `PREREGISTER` stage, the hypothesis, model version, initial state, parameters, input influences, control conditions, primary and secondary metrics, termination criteria, threshold values, random seed when applicable, and failure conditions are specified. The `FREEZE` operation fixes this specification before the experiment begins. Once the process proceeds to `RUN`, parameters, metrics, and criteria must not be adjusted in order to obtain a desired result. If a flaw in the experimental specification is identified, the experiment is terminated, the reason is recorded in `TRACE`, and the modified experiment is registered as a new version.

The `AUDIT` stage provides independent verification of quantitative results against the frozen specification. Numerical auditing involves reproducing the main values, checking units, indexing, normalization, initial conditions, threshold values, and consistency with the current model version, as well as performing sensitivity analysis when necessary. The audit result may have the status `VERIFIED_NUMERICALLY`, `PROVISIONAL`, `FAILED_REPRODUCTION`, or `AMBIGUOUS_SPECIFICATION`; final quantitative claims are permitted only for properly verified results.

Importantly, HAAG does not equate research success with confirmation of the initial hypothesis. Negative and null results are valid research outcomes:

Negative Results $\in$ *Results*, *Null Results* $\in$ *Results*.

A failed experiment is likewise not removed from the research history but is recorded as a distinct state, together with the cause of failure, the information obtained, the claims that it refutes or calls into question, and the possible subsequent transitions. Modifying a model or experimental procedure after a negative result requires a new justified transition and, when necessary, renewed preregistration.

This approach is complemented by a mechanism for tracing scientific claims. For each consequential claim c_i , a record is maintained specifying its type, evidential basis, current status, and permissible strength of formulation. The sources of support may include

Support(c_i) $\in$ {*Equation*, *Experiment*, *Source*, *Dataset*, *HumanDecision*}.

A claim may have the status SUPPORTED, PARTIALLY_SUPPORTED, UNSUPPORTED, CONTRADICTED, UNVERIFIABLE, or PROVISIONAL. A fundamental constraint is that the strength of the final formulation must not exceed the strength of its verified support.

Thus, for the set of scientific claims, the following mapping is established:

Claim $\rightarrow$ *Provenance*,

linking each significant element of scientific argumentation to the basis on which it was introduced. This extends the traditional traceability of bibliographic sources to the traceability of scientific argumentation, since a claim may be supported not only by an external source but also by a mathematical derivation, an experiment, a dataset, or a recorded researcher decision.

Verification of results is directly related to the architectural separation of three layers in HAAG:

Research Layer $\rightarrow$ *Writing Layer* $\rightarrow$ *Artifact Layer*.

The *Research Layer* encompasses problem formulation, hypotheses, models, evidence, experiments, falsification, and interpretation. The *Writing Layer* transforms the accepted and verified research state into structured scientific text. The *Artifact Layer* generates research-related figures, tables, prompts, result-provenance data, and reproducibility materials. The HAAG specification requires that the demands of writing or artifact preparation must not alter research decisions or the interpretation of results.

This principle can be expressed in general form as

Research → Writing,

but not as the reverse influence

Writing ↗ Research.

Thus, the need to fill a particular structural element of an article does not justify creating a missing result. If the corresponding stage of the research lacks sufficient evidential support, HAAG must record an incomplete or unsupported state rather than compensate for it by generating plausible text.

For articles (`ARTICLE`) and short scientific papers (`SHORT_PAPER`), HAAG uses IMRaD as the basic structure. However, this structure defines the form of presentation rather than the logic by which the results are obtained. The `Results` section must contain only results that were actually obtained, including negative and null results, while the conclusions must be limited to claims supported by the results. The formal logic of the publication is represented by the chain

Gap → Objective → Method → Result → Interpretation → Conclusion.

In terms of the state model, the scientific text can be represented as a projection of the verified final research state:

$$P = \Pi_W(S_n^{\text{verified}}),$$

where Π_W is the scientific writing operator that transforms the verified components of S_n into a structured manuscript. An important property of HAAG follows from this formulation: the language model must not fill a missing link in the argument merely to ensure textual coherence. Such a gap must instead be identified as an incomplete or insufficiently verified research state, requiring a return to the corresponding stage of the research process.

The publication itself is only one of the outputs produced by HAAG. The final package may also include figure specifications and prompts for their generation, a hypothesis tree, a history of agent actions, human decisions, rejected branches, failed experiments, an experiment registry, claim and source provenance, working search queries, numerical audit results, unresolved problems, and a reproducibility package.

The output of the framework should therefore be defined more broadly:

Research Output = Publication + Evidence + Trace + Artifacts + Reproducibility.

In this model, the publication is a compact communication form of a broader research output, while `Evidence`, `Trace`, `Artifacts`, and `Reproducibility` make it possible to verify the provenance of

claims, reconstruct the research trajectory, and, to the extent permitted by the nature of the study, reproduce the obtained results.

The practical application of HAAG was particularly important in shaping this architecture. Version v1.1 was extended relative to v1.0 on the basis of experience gained from an end-to-end theoretical and modeling research workflow. The specification explicitly states that this experience led to the inclusion of mechanisms for preregistered experiments, negative results, numerical auditing, claim provenance, working Scholar queries, prompts for generating scientific figures, and a GitHub-ready reproducibility package.

The evolution from HAAG v1.0 to v1.1 can therefore be regarded as an initial iteration of reflexive framework improvement: practical execution of the research cycle is used not only to obtain a domain-specific result but also to identify requirements for the methodology itself. At the same time, such practical application demonstrates the feasibility of the approach but does not, by itself, constitute a comparative empirical evaluation of its effectiveness. Such an evaluation would require separate experiments with predefined criteria, baseline approaches, and repeatable application scenario.

6. Discussion and Conclusions

The proposed approach does not imply the autonomization of scientific inquiry or the transfer of scientific responsibility to a language model. In HAAG, the researcher remains the bearer of scientific intent and the subject responsible for fundamental decisions, while the agentic system extends the operational capabilities of the research process. Specialized agents can formulate alternative hypotheses in parallel, identify contradictions, search for falsification conditions, design experiments, and verify sources and numerical results; however, the selection of a scientifically consequential trajectory remains linked to explicit decisions by the researcher. This division of functions reflects the fundamental HAAG principle according to which agents explore the space of possibilities, while the human shapes the research trajectory.

In this sense, HAAG occupies an intermediate position between two limiting models of scientific work:

Human does everything

and

Autonomous AI does research.

The proposed model is based on their functional integration:

**HumanScientific Judgment + Agentic Research Capabilities +
+ Verification + Traceability + Reproducibility.**

The agentic component is used primarily to broaden the scope of research exploration and to systematically evaluate alternatives, while the human defines the scientific objective, assesses the substantive relevance of proposed directions, and controls transitions capable of changing the problem

formulation, method, evidence base, or interpretation of results. This approach makes it possible to exploit the productivity and variability of LLMs without equating agent autonomy with autonomy of scientific decision-making.

Summarizing the architecture of the framework, HAAG can be regarded as a prompt-native, domain-specific, no-code language for interactive scientific research. Natural language provides the semantics of research operations; logical primitives govern states and transitions; specialized LLM agents implement distributed research functions; external scientific sources and computational tools provide informational and numerical support; provenance mechanisms maintain the traceability of claims and decisions; and the `HUMAN` operator ensures researcher involvement in scientifically consequential transitions. This conceptual interpretation of HAAG is embedded in the v1.1 specification.

An important property of this model is the change in the criterion of research progress itself. In a conventional text-generation system, this criterion may effectively be the appearance of the next semantically coherent fragment of the manuscript. In HAAG, transition to the next state must be determined not by the needs of the text but by scientific justification. Therefore, the fundamental question of the framework is formulated as

What scientifically justified research transition should occur next?]

rather than

What text should be generated next?

This principle is explicitly defined in HAAG v1.1 and is implemented through the extended cycle `THINK → CHALLENGE → VERIFY → DECIDE → PREREGISTER → RUN → AUDIT → TRANSITION → WRITE → FIGURE → VERIFY SOURCES → CITE → PACKAGE → FINAL AUDIT`.

This also leads to a fundamental distinction between the output of HAAG and conventional LLM-assisted manuscript preparation. The final object is not merely the publication text but the verified research state together with its evidence, decision history, rejected alternatives, experimental results, claim provenance, scientific artifacts, and reproducibility materials. The scientific publication is a compact representation of this state but does not exhaust it.

In compact form, the proposed paradigm can be represented by the sequence

**Topic+Profile → Research Program → Human-Agent Research →Trajectory →
→ Verified Results → Verified Scientific Publication + Reproducibility Package.**

This scheme summarizes the main components of HAAG: the initial research intent is compiled into a research program; execution of this program produces a human-agent research trajectory; the resulting findings undergo verification and auditing; and only then are they transformed into a scientific publication and the associated reproducibility package.

At the same time, the current practical evaluation of HAAG does not justify treating the effectiveness of the framework as universally established. The v1.1 specification was developed, in particular, on the basis of experience gained from an end-to-end theoretical and modeling research cycle, which made it possible to refine the mechanisms for preregistration, numerical auditing, handling negative

results, provenance, and reproducibility. However, demonstrating feasibility in a single scenario should be complemented by systematic experimental evaluation.

A promising direction for future work is to compare HAAG with conventional LLM-assisted research workflows and other agentic approaches across different types of scientific tasks, domains, and models. Such comparisons could examine the proportion of unsupported claims, the number of errors detected during internal verification, reproducibility of numerical results, completeness of provenance, the number and nature of substantive `HUMAN` transitions, the proportion of preserved negative results, and the ability of an independent researcher to reconstruct the path from the initial hypothesis to the final conclusion. These indicators should be regarded as a program for further validation of the framework rather than as already established measures of its effectiveness.

Thus, this work proposes and formalizes Human-Agentic Article Genesis as a human-agentic framework for organizing scientific research and creating publications. Its central object is the research trajectory, rather than the process of text generation. The formal state model, specialized agent society, `HUMAN` operator, branching and backtracking mechanisms, preregistration and numerical auditing, claim provenance, handling of sources and negative results, and construction of the reproducibility package form an integrated system in which writing a scientific article is a consequence of research that has been conducted and verified.

The central idea of HAAG can therefore be defined as a transition from LLM-assisted generation of scientific text to human-agentic execution of a traceable scientific research process. In this formulation, the generative model ceases to function as a direct “article generator” and instead becomes a component of a formalized research environment in which agentic capabilities are combined with human scientific judgment, while the final publication is generated from the results of a traceable and verified research trajectory.

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